

AI ASSISTED CODING

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BATCH – 03

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ASSIGNMENT – 13.2

LAB – 13 : Code Refactoring – Improving Legacy Code with AI Suggestions.

Task – 01 : Refactoring – Eliminating Repetitive Logic.

Prompt : Use AI to analyze the Python code and identify repeated calculations for area and perimeter. Refactor the code by creating a reusable function with proper docstrings while ensuring the output remains the same.

Original Code & Output :



The screenshot shows a code editor window titled 'TASK 01'. It contains a section labeled 'ORIGINAL CODE' with the following Python code:

```
[1] length = 8
width = 4
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))

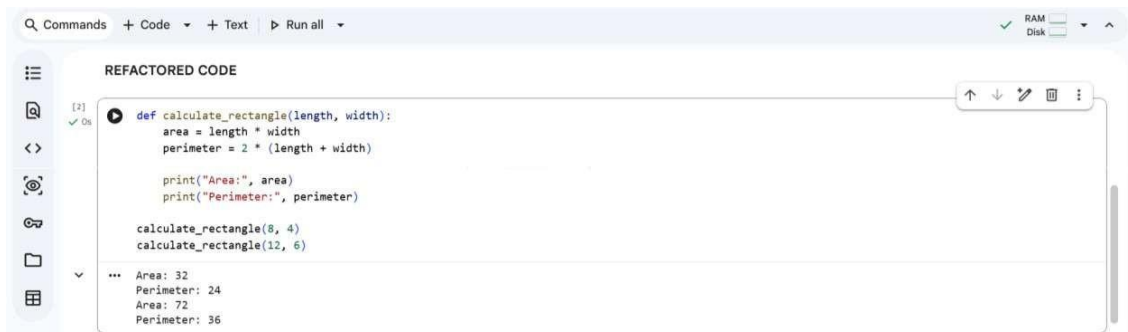
length = 12
width = 6
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))
```

Below the code, the output is displayed:

```
... Area: 32
Perimeter: 24
Area: 72
Perimeter: 36
```

The editor interface includes a sidebar with icons for file explorer, search, and other tools. The bottom status bar shows 'Variables', 'Terminal', '1:30 PM', and 'Python 3'.

Refactored Code & Output :



The screenshot shows the same code editor window, now displaying the 'REFACTORED CODE' section. The code is as follows:

```
[2] def calculate_rectangle(length, width):
    area = length * width
    perimeter = 2 * (length + width)

    print("Area:", area)
    print("Perimeter:", perimeter)

calculate_rectangle(8, 4)
calculate_rectangle(12, 6)
```

The output remains the same as in the previous screenshot:

```
... Area: 32
Perimeter: 24
Area: 72
Perimeter: 36
```

The editor interface is consistent with the previous screenshot, showing the same sidebar and status bar.

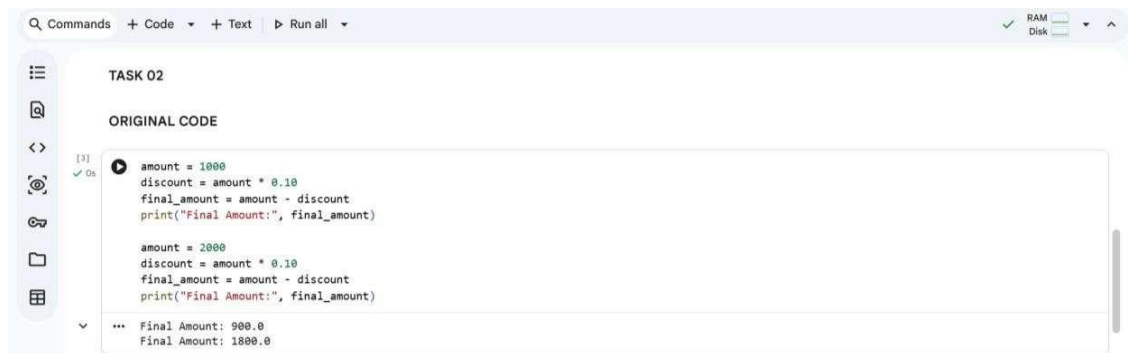
Explanation :

The repeated calculations for area and perimeter were moved into a reusable function called `calculate_rectangle`. This improves code readability and reduces duplication. The program still produces the same output but is easier to maintain.

Task – 02 : Refactoring – Modularizing Inline Calculations.

Prompt : Use AI to refactor the Python code by identifying repeated discount calculations and converting them into a reusable function with proper documentation.

Original Code & Output :



```
TASK 02

ORIGINAL CODE

[3] ✓ Os
1 amount = 1000
2 discount = amount * 0.10
3 final_amount = amount - discount
4 print("Final Amount:", final_amount)

5
6 amount = 2000
7 discount = amount * 0.10
8 final_amount = amount - discount
9 print("Final Amount:", final_amount)

10
11 ... Final Amount: 900.0
12 ... Final Amount: 1800.0
```

Refactored Code & Output :



```
REFACTORED CODE

[4]
1 def calculate_final_amount(amount):
2     discount = amount * 0.10
3     final_amount = amount - discount
4     print("Final Amount:", final_amount)
5
6 calculate_final_amount(1000)
7 calculate_final_amount(2000)
8
9 ... Final Amount: 900.0
10 ... Final Amount: 1800.0
```

Explanation :

The repeated discount calculation logic was moved into a function called `calculate_final_amount`. This makes the code modular and reusable for different input values. It also improves readability and reduces redundant code.

Task – 03 : Refactoring – Improving Loop and Conditional Performance.

Prompt : Use AI to optimize the Python program by replacing inefficient nested loops with a faster lookup structure while keeping the output unchanged.

Original Code & Output :



```
Q Commands + Code + Text ▶ Run all ✓ RAM Disk
```

TASK 03

ORIGINAL CODE

```
[5] ✓ Os
1 students = ["Anil", "Bhavya", "Charan", "Divya"]
2 check_list = ["Charan", "Esha", "Bhavya"]
3 for name in check_list:
4     exists = False
5     for student in students:
6         if name == student:
7             exists = True
8     if exists:
9         print(name, "found")
10    else:
11        print(name, "not found")
... Bhavya found
```

Refactored Code & Output :



```
Q Commands + Code + Text ▶ Run all ✓ RAM Disk
```

REFACTORED CODE

```
[6] ✓ Os
1 students = ["Anil", "Bhavya", "Charan", "Divya"]
2 check_list = ["Charan", "Esha", "Bhavya"]
3 student_set = set(students)
4 for name in check_list:
5     if name in student_set:
6         print(name, "found")
7     else:
8         print(name, "not found")
... Charan found
... Esha not found
... Bhavya found
```

Explanation :

The nested loops were replaced with a set data structure, which provides faster lookup operations. Instead of checking each student repeatedly, membership is checked directly using `in`. This improves performance from $O(n^2)$ to $O(n)$.

Task – 04 : Refactoring – Replacing Hardcoded Values with Constants.

Prompt : Use AI to identify hardcoded numerical values in the simple interest calculation and replace them with descriptive constants to improve maintainability.

Original/Refactored Code & Output :

```
Q Commands + Code + Text ▶ Run all ✓ RAM Disk ^
```

TASK 04

ORIGINAL CODE

```
[7] ✓ 0s  
print("Simple Interest:", (1000 * 2 * 5) / 100)  
print("Simple Interest:", (2000 * 2 * 5) / 100)  
  
Simple Interest: 100.0  
Simple Interest: 200.0
```

REFACTORED CODE

```
[8] ✓ 0s  
RATE = 2  
TIME = 5  
PERCENT = 100  
def calculate_simple_interest(principal):  
    interest = (principal * RATE * TIME) / PERCENT  
    print("Simple Interest:", interest)  
    calculate_simple_interest(1000)  
    calculate_simple_interest(2000)  
  
... Simple Interest: 100.0  
Simple Interest: 200.0
```

⌵ Variables Terminal 1:48 PM Python 3

Explanation :

Hardcoded values such as rate, time, and percentage were replaced with named constants (RATE, TIME, PERCENT). This makes the code easier to understand and modify. If the rate or time changes, it only needs to be updated once.

Task – 05 : Refactoring – Enhancing Variable Naming Code Clarity.

Prompt : Use AI to improve code readability by replacing unclear variable names with meaningful ones and adding comments explaining the logic.

Original/Refactored Code & Output :

```
Q Commands + Code + Text ▶ Run all ✓ RAM Disk ^
```

ORIGINAL CODE

```
[9] ✓ 0s  
x = 50  
y = 30  
z = x + y  
print(z)  
  
80
```

REFACTORED CODE

```
[10] ✓ 0s  
# Define two numbers  
num1 = 50  
num2 = 30  
  
# Calculate the sum of the numbers  
total_sum = num1 + num2  
  
# Display the result  
print(total_sum)  
  
... 80
```

⌵ Variables Terminal 1:52 PM Python 3

Explanation :

The variables x, y, and z were renamed to num1, num2, and total_sum to improve clarity. Comments were added to explain each step of the code. This makes the program easier to understand for other developers.

THANK YOU!!